Mohamed Rabie Ghediri

Data Engineering Student

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Profile

Final-year Data Engineering and Decision Systems engineering student specializing in data pipelines, distributed processing, cloud platforms, analytics engineering, and geospatial data systems. Experienced in building reliable ETL/ELT workflows, streaming architectures, data quality frameworks, and analytics-ready platforms using modern data engineering tools. Seeking a final-year project internship in Data Engineering, Cloud Data Platforms, Analytics Engineering, or Big Data.

Experience

Yonnov'IA – AI / Data Science Intern Marseille, France
Python, FastAPI, React, TypeScript, PostgreSQL, Sentence Transformers, Statsmodels June – August 2026

- Built an end-to-end platform for scientific-domain discovery, historical publication analysis, and trend forecasting through 2030, supporting OpenAlex and local CSV datasets.
- Implemented E5 semantic discovery to construct research families and identify terminology variants from publication data.
- Developed a forecasting engine comparing Naive, Linear, Holt, ETS, and ARIMA models with rolling-origin backtesting (MAE, RMSE, MASE, sMAPE) for automatic model selection.

Otospex Solutions – Data Engineering Intern Djerba, Tunisia
Python, SQL, Airflow, dbt, PostgreSQL, Docker, Power BI June – August 2025

- Built an Airflow ELT pipeline loading multi-source vehicle data into a PostgreSQL warehouse.
- Built incremental dbt Bronze/Silver/Gold layers for clean, standardized star-schema data.
- Implemented automated data quality checks for completeness, validity, uniqueness, and consistency on vehicle listings.

Projects

Real-Time Flight Delay Prediction System
Kafka, Spark Structured Streaming, Cassandra, MLflow, FastAPI, Grafana, Python

Built a real-time data processing pipeline using Kafka and Spark Structured Streaming to process and analyze **1M+ flight records**. Integrated Cassandra for data storage and MLflow for model tracking and experiment management. Developed monitoring dashboards with Grafana and implemented a predictive component achieving an AUC between **0.78 and 0.84**.

AutoMarket Intel – Automotive Market Data Platform
Python, DuckDB, dbt, AWS S3, Snowflake, Parquet, Terraform, GitHub Actions

Built a data engineering pipeline to collect, transform, and organize **108K+ automotive listings** into an analytics-ready data platform. Used DuckDB and dbt to implement data transformations and validation, with Parquet and AWS S3 for data storage. Implemented **155 data tests** and automated infrastructure and CI/CD workflows using Terraform and GitHub Actions.

CropWatch – Geospatial Data Engineering & Agricultural Intelligence Platform
Python, Dagster, MinIO, Rasterio, Xarray, GeoPandas, Parquet, PostGIS, Sentinel-2, CHIRPS, ERA5-Land

Designing a regional geospatial data platform to ingest and organize satellite imagery, rainfall, climate, and soil datasets for agricultural monitoring. Planning data pipelines for raster processing, spatial transformations, temporal alignment, and regional feature aggregation. Structuring processed geospatial datasets into analytics-ready formats to support drought-stress monitoring and downstream machine learning workflows.

Technical Skills

Programming Languages: Python, SQL, Java, Scala, JavaScript/TypeScript
AI & Generative AI: LLMs, RAG, NLP, Semantic Search, Hugging Face, LangChain, LangGraph, FAISS
Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, MLflow
Data Engineering: Apache Kafka, Apache Spark, Apache Airflow, Databricks, Hadoop, Docker, ETL/ELT Pipelines
Databases & Search: PostgreSQL, MongoDB, Cassandra, Elasticsearch, Vector Databases
Frameworks & Tools: FastAPI, React, Vue.js, Git, Power BI
Cloud & Deployment: AWS S3, Docker, Kubernetes, GitHub Actions
Languages: Arabic (Native), English (Advanced), French (Intermediate), German (Beginner)

Education

Engineering Degree – Data Engineering & Decision Systems (IDSD) 2024 – Present
ENET'Com, Sfax

Engineering Preparatory Cycle – Physics & Technology (PT) 2021–2024
IPEIN, Nabeul